

Project Report

Title: AI-Based Story Generator Using GPT-2 and Gradio

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Abstract

This project utilizes the GPT-2 language model from Hugging Face's transformers library to generate creative and engaging stories based on a given character name, personality traits, and story theme. The application provides an intuitive web interface built with Gradio, allowing users to input details like character name, personality, and theme. The system then generates a personalized story, which can be further explored by users. This AI-powered story generator can be applied in areas like creative writing, gaming, and interactive fiction.

Introduction

The goal of this project is to create an AI-based system capable of generating personalized stories. The system is powered by GPT-2, a popular transformer-based model that excels at text generation. By using character-driven inputs, the model creates a narrative that follows the inputted character's traits and chosen story theme.

Incorporating Gradio allows for a user-friendly interface, enabling easy interaction with the AI model without needing to write any code. Users can see immediate results of their inputs in a generated text format.

Tools and Technologies Used

- **Python:** The primary programming language used for building the system.

- **Transformers Library:** A library from Hugging Face, which provides access to pre-trained models such as GPT-2.
- **GPT-2 Model:** A pre-trained transformer-based model for natural language processing (NLP), used here for text generation.
- **Gradio:** A Python library used to create interactive interfaces for machine learning models, enabling easy deployment of the application in a web browser.
- **Pip:** Used for installing the required libraries (`transformers` and `gradio`).

OBJECTIVE

To develop a document-based question-answering tool that runs in Google Colab.

To use a lightweight embedding and language generation model suitable for Colab's CPU/GPU environment.

To make research and learning easier by retrieving answers from uploaded PDFs.

METHODOLOGY

Step-by-Step Flow in Google Colab:

Step 1: Install Required Libraries

```
!pip install transformers gradio
```

Step 2: Load a Pre-trained Language Model

Using GPT-2 via Hugging Face's transformers.

```
from transformers import pipeline, set_seed

# Load text generation pipeline
generator = pipeline('text-generation', model='gpt2')
set_seed(42)
```

Step 3: Define the Prompt-Based Story Generator Function

```
def generate_story_model(character_name, character_personality, story_theme):
    # Prompt engineering: Guide the model with a detailed context
    prompt = (
        f"Write a captivating {story_theme.lower()} story. "
        f"The story follows {character_name}, who is {character_personality}. "
        f"The adventure begins like this:\n"
    )

    # Generate story
    story = generator(prompt, max_length=500, num_return_sequences=1,
do_sample=True, temperature=0.9)[0]['generated_text']
    return story
```

Step 4: Build a Gradio Web Interface:

```
import gradio as gr

def launch_gradio():
    interface = gr.Interface(
        fn=generate_story_model,
        inputs=[
            gr.Textbox(label="Character Name"),
            gr.Textbox(label="Personality Traits"),
            gr.Textbox(label="Story Theme (e.g., Fantasy, Romance, Mystery)")
        ],
        outputs="text",
        title="AI Story Generator",
        description="Generate stories based on a character name, personality, and theme using GPT-2."
```

```
)  
return interface  
  
interface = launch_gradio()  
interface.launch(share=True)
```

SAMPLE OUTPUT:

835c39f85428107406.gradio.live

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AI Story Generator

Generate stories based on a character name, personality, and theme using GPT-2.

Character Name

DHINESH

Personality Traits

KIND

Story Theme (e.g., Fantasy, Romance, Mystery)

MYSTERY

Clear

Submit

output

Write a captivating mystery story. The story follows DHINESH, who is KIND. The adventure begins like this:

A dark-horse adventure with great tension. You are in love with your parents, and yet you will find yourself in a dangerous place.

A very dark, strange situation. There are clues leading to your family's deaths.

A mysterious man is holding you hostage in dark, strange circumstances.

A boy is the heir to the throne of your family. His name is HINESH. But you refuse to recognize him.

You are in the middle of a war that is the ultimate danger. The people of the universe are not aware of the danger.

No matter how much you ask for mercy and protection, there are only too many who know the truth.

The world is not safe.

Flag

Conclusion

This AI-based story generator provides an interactive and engaging way to create personalized narratives based on user inputs. By leveraging GPT-2 for text generation and Gradio for a user-friendly interface, this project showcases the power of AI in the field of creative writing. The system can be extended to support more customization options, such as setting specific plot structures or including multiple characters. With further improvements and optimizations, it could serve as a valuable tool for writers, educators, and anyone interested in creative storytelling.

References

- Hugging Face: <https://huggingface.co>
- SentenceTransformers: <https://www.sbert.net>
- Google Colab: <https://colab.research.google.com>